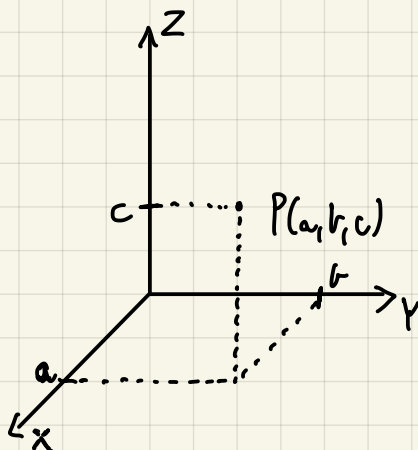
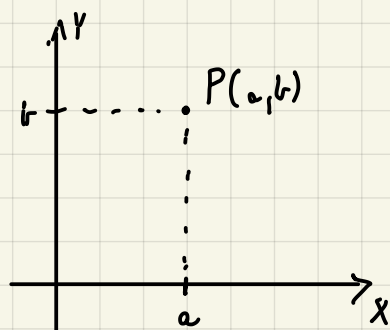
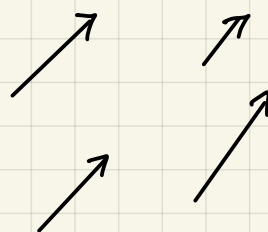


Przestrzeń $\mathbb{R}^2$ a $\mathbb{R}^3$



$B = (b_x, b_y, b_z)$
 $\vec{v} = \vec{AB} = [b_x - a_x, b_y - a_y, b_z - a_z]$
 $|\vec{v}| = \sqrt{(b_x - a_x)^2 + (b_y - a_y)^2 + (b_z - a_z)^2}$

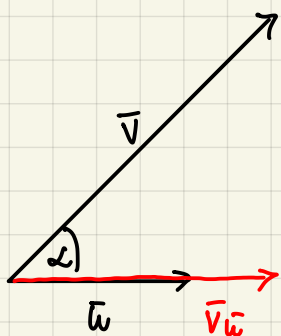
Wektor swobodny



$$\vec{0} = [0, 0, 0]$$

$$\vec{u} \cdot \vec{v} = \vec{u} \circ \vec{v} = \vec{u} \vec{v} \stackrel{\text{def}}{=} |\vec{u}| \cdot |\vec{v}| \cdot \cos(\angle(\vec{u}, \vec{v})) \stackrel{\text{tw}}{=} u_x v_x + u_y v_y + u_z v_z$$

$$\vec{a} \perp \vec{b} \Leftrightarrow \vec{a} \cdot \vec{b} = 0$$



$$\vec{v}_u = v_u \cdot \frac{\vec{u}}{|\vec{u}|} = \left(\frac{\vec{u} \cdot \vec{v}}{|\vec{u}|^2} \right) \cdot \vec{u}$$

$$|\vec{v}_u| = v \cos \alpha$$

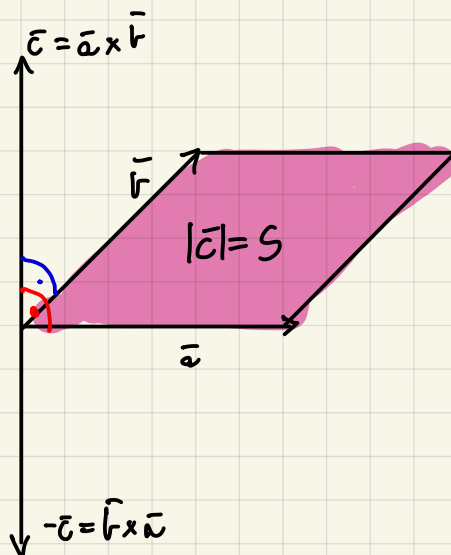
Iloczyn wektorowy

Iloczyn wektorowy wektorów $\vec{a}$ i $\vec{b}$ jest to wektor $\vec{c} = \vec{a} \times \vec{b}$, że

(i) $\vec{c} \perp \vec{a}$ i $\vec{c} \perp \vec{b}$

(ii) wektory $\vec{a}, \vec{b}, \vec{c}$ tworzą układ prawoskalarny

(iii) $|\vec{c}| = |\vec{a}| \cdot |\vec{b}| \cdot \sin \angle(\vec{a}, \vec{b})$



Wnioski (własności iloczynu wektorowego)

(i) $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$

(ii) $|\vec{a} \times \vec{b}|$ jest równy polu równoległoboku rozpiętego na wektorach $\vec{a}$ i $\vec{b}$

(iii) $\vec{a} \parallel \vec{b} \Leftrightarrow \vec{a} \times \vec{b} = [0, 0, 0]$

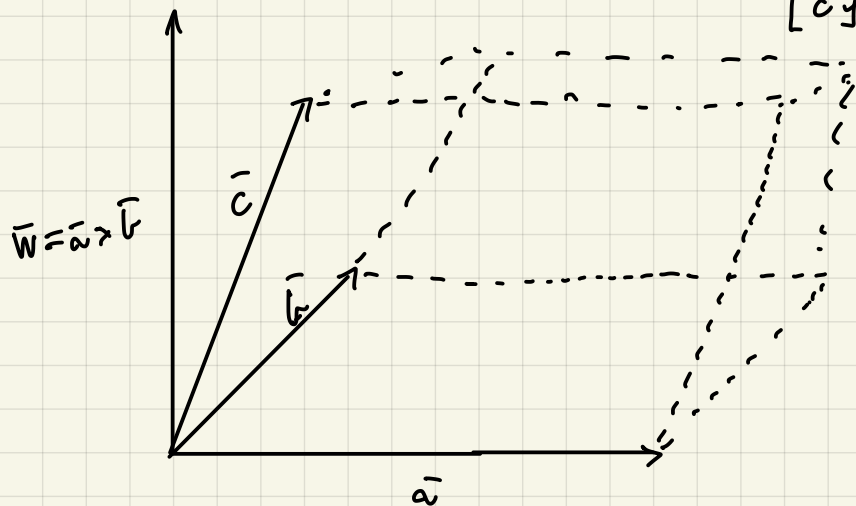
Twierdzenie

Iloczyn wektorowy obliczamy według wzoru

$$\vec{u} \times \vec{v} = \left[\begin{bmatrix} u_y & u_z \\ v_y & v_z \end{bmatrix}, -\begin{bmatrix} u_x & u_z \\ v_x & v_z \end{bmatrix}, \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix} \right]$$

Iloczyn mieszany trzech wektorów

$$W(\vec{a}, \vec{b}, \vec{c}) = \vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} = \det \begin{bmatrix} \vec{a} \\ \vec{b} \\ \vec{c} \end{bmatrix} = \det \begin{bmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{bmatrix} \in \mathbb{R}$$



Wnioski

(i) $W(\vec{a}, \vec{b}, \vec{c}) = 0$, jeżeli wektory $\vec{a}, \vec{b}$ i $\vec{c}$ są współliniowe.

(ii)

$$\cos \angle(\vec{u}, \vec{v}) = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|}$$

$$\sin \angle(\vec{u}, \vec{v}) = \frac{|\vec{u} \times \vec{v}|}{|\vec{u}| \cdot |\vec{v}|}$$

$$\tan \angle(\vec{u}, \vec{v}) = \frac{|\vec{u} \times \vec{v}|}{|\vec{u} \cdot \vec{v}|}$$

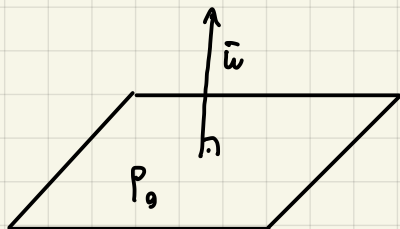
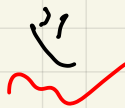
Równanie ogólne płaszczyzny

$$H: Ax + By + Cz + D = 0$$

$$A^2 + B^2 + C^2 > 0$$

Płaszczyzna przez punkt i prostopadła do wektora

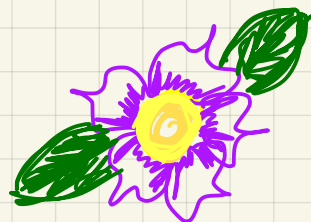
$$P_0 = (x_0, y_0, z_0) \quad \vec{n} = [A, B, C]$$



$$H: A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

$$l: \begin{cases} x = x_0 + k_x t \\ y = y_0 + k_y t \\ z = z_0 + k_z t \end{cases}$$

Milego ☺

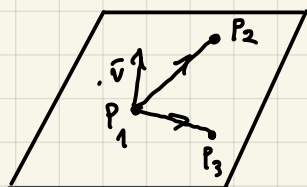


Książek
for u ☺

$$P_1 = (-2, 4, 1)$$

$$P_3 = (2, 0, -1)$$

$$P_2 = (0, 2, -1)$$



$$\overline{P_1 P_2} = [2, -2, -2] = 2 \overbrace{[1, -1, -1]}^{\bar{u}}$$

$$\overline{P_1 P_3} = [4, -4, -2] = 2 \underbrace{[2, -2, -1]}_{\bar{v}}$$

$$\bar{u} \times \bar{v} = \begin{bmatrix} 1 & -1 & -1 \\ 2 & -2 & -1 \end{bmatrix} \\ \hline [-1, -1, 0] = -1[1, 1, 0]$$

$$H \perp \hat{n} = [1, 1, 0]$$

$$H: 1(x - (-2)) +$$